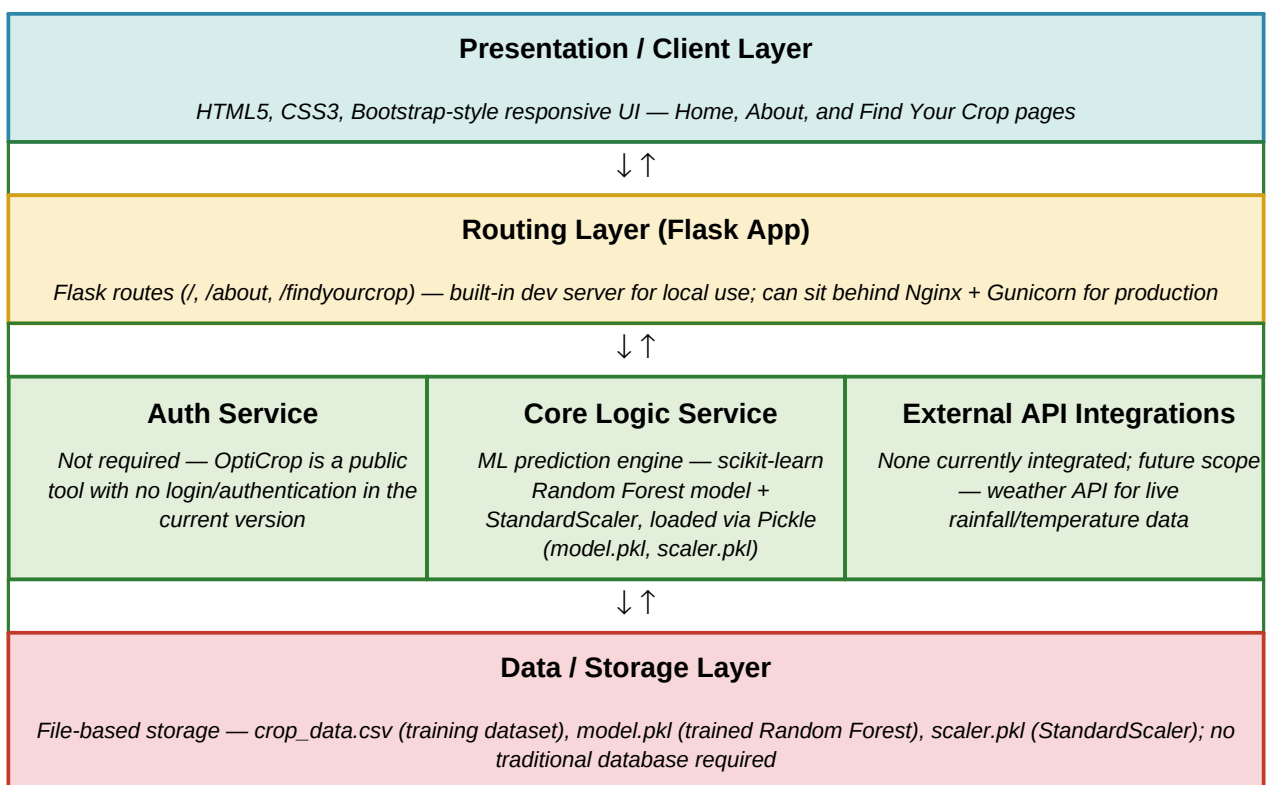


Solution Architecture

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of OptiCrop's architecture across the Presentation, Application, and Data layers.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the Home, About, and Find Your Crop pages; collects N, P, K, temperature, humidity, pH, and rainfall from the user and displays the predicted crop.	HTML5, CSS3, Jinja2 templates
Routing Layer (App Server)	Handles HTTP requests, routes them to the right page/logic, and returns rendered templates or prediction results.	Flask (Python), Werkzeug dev server

Component Name	Description / Role in Architecture	Technologies Used
Core Logic / ML Service	Loads the trained model and scaler, validates and scales incoming input, and generates the crop prediction.	Python, scikit-learn (Random Forest, KNN, Decision Tree, Logistic Regression, K-Means), Pickle
Data / Storage Layer	Stores the training dataset and the persisted model/scaler artifacts used at prediction time.	CSV file (crop_data.csv), Pickle files (model.pkl, scaler.pkl)